

99.500 Applied Mathematics for Engineering Spring Semester AY2026

Modeling & Nondimensionalization

Week 2 Outline

- Flow Reactor
- Axial Dispersion in Flow Reactor
- Batch Reactor
- Heat Transfer Semi-infinite block

For the flow reactor problem, we derived and discussed last week, how well do you understand it? Please rate your understanding on a scale of 1 to 5, where 1 means 'not understood at all' and 5 means 'fully understood'.

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



Multiple Choice

Flow Reactor

Governing equation

$$\begin{cases} \frac{dC_A}{dx} = -\frac{k}{u}C_A \\ C_A(x=0) = C_{A,0} \end{cases}$$

Boundary condition

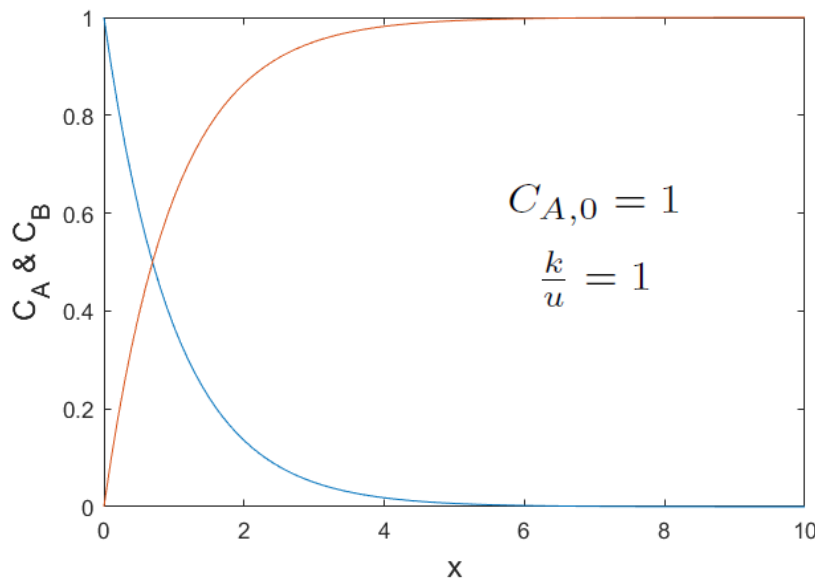
Separation of variables

$$\frac{dC_A}{C_A} = -\frac{k}{u}dx \Rightarrow \int \frac{dC_A}{C_A} = \int -\frac{k}{u}dx$$

$$\ln C_A = -\frac{k}{u}x + c_1 \Rightarrow C_A = c_2 e^{-\frac{k}{u}x}$$

$$C_A(x=0) = C_{A,0} \Rightarrow c_2 = C_{A,0}$$

$$C_A = C_{A,0}e^{-\frac{k}{u}x} \quad \& \quad C_B = C_{A,0}(1 - e^{-\frac{k}{u}x})$$



What insights can we derive from the graph?

What is Nondimensionalization?

A mathematical technique used to remove physical units from equations by introducing **dimensionless variables**.

- Simplifies equations
- Identifies key parameters
- Generalizes solutions
- Improves numerical stability

Steps of Nondimensionalization

1. Identify characteristic scales for variables
2. Define dimensionless variables using these characteristic scales
3. Rewrite the original equation in terms of these new variables
4. Identify key dimensionless groups, which determine the system's behaviour

Flow Reactor : Nondimensionalization

$$\text{Let } \tilde{C}_A = \frac{C_A}{C_{A,0}} \text{ \& } \tilde{x} = \frac{x}{L}$$

The governing and BC equation is its dimensionless form:

$$\frac{d\tilde{C}_A}{d\tilde{x}} = -\alpha\tilde{C}_A \text{ where } \alpha = \frac{kL}{u}$$

$$\tilde{C}_A(\tilde{x} = 0) = 1$$

$$\alpha = \frac{kL}{u} = \frac{kC_{A,0}LA}{uC_{A,0}A}$$

$$\begin{aligned} &\sim \frac{\text{number of moles of A consumed in the reactor tube per unit time}}{\text{number of moles of A flowing through the reactor tube per unit time}} \\ &\sim \frac{\text{reaction rate for reactant A}}{\text{flow rate for reactant A}} \end{aligned}$$

Flow Reactor: Result and Interpretation

Solve the dimensionless ODE:

$$\frac{d\tilde{C}_A}{d\tilde{x}} = -\alpha\tilde{C}_A \Rightarrow \frac{d\tilde{C}_A}{\tilde{C}_A} = -\alpha d\tilde{x}$$

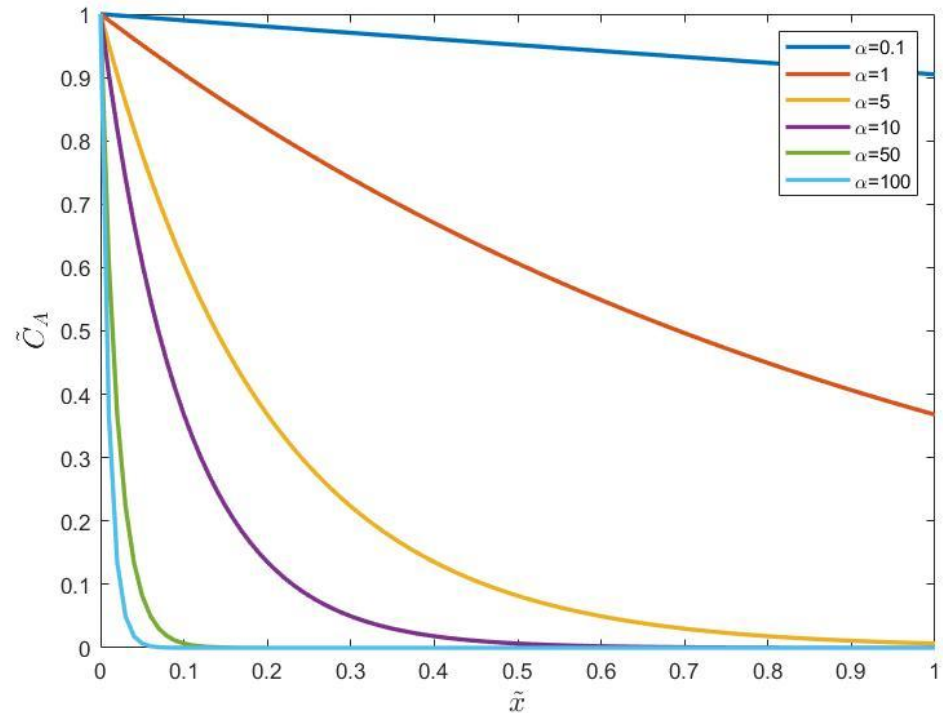
$$\ln \tilde{C}_A = -\alpha\tilde{x} + c_1 \Rightarrow \tilde{C}_A = c_2 e^{-\alpha\tilde{x}}$$

$$\tilde{C}_A(\tilde{x} = 0) = 1 \Rightarrow c_2 = 1 \Rightarrow \tilde{C}_A = e^{-\alpha\tilde{x}}$$

How about B?

What insights can we derive from the graph?

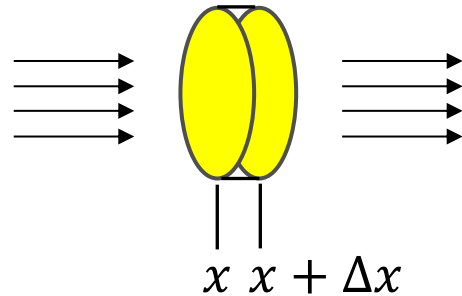
$$\alpha = \frac{kL}{u} = \frac{kC_{A,0}LA}{uC_{A,0}A}$$



Axial Dispersion (AD)

Flow reactor with invalid assumption

Dispersion term: proportional to concentration gradient



In: $uC_A|_x A - D \left. \frac{dC_A}{dx} \right|_x A$

Out: $uC_A|_{x+\Delta x} A - D \left. \frac{dC_A}{dx} \right|_{x+\Delta x} A$

Generation: 0

Consumption: $-r_A \Delta V = kC_A A \Delta x$

Accumulation: 0 for steady-state

Shell balance equation:

$$0 = \left[uC_A|_x A - D \left. \frac{dC_A}{dx} \right|_x A \right] - \left[uC_A|_{x+\Delta x} A - D \left. \frac{dC_A}{dx} \right|_{x+\Delta x} A \right] - kC_A A \Delta x$$

With re-arrangement:

$$D \frac{\partial^2 C_A}{\partial x^2} - u \frac{\partial C_A}{\partial x} - kC_A = 0$$

Boundary conditions:

$$uC_{A,0} = \left[uC_A - D \frac{dC_A}{dx} \right]_{x=0} \text{ \& } \left[\frac{dC_A}{dx} \right]_{x=L} = 0 \text{ (Danckwerts BC)}$$

AD: Nondimensionalization

$$\text{Let } \tilde{C}_A = \frac{C_A}{C_{A,0}} \text{ \& } \tilde{x} = \frac{x}{L}$$

$$\Rightarrow \begin{cases} D \frac{\partial^2 (\tilde{C}_A C_{A,0})}{\partial (\tilde{x} L)^2} - u \frac{\partial (\tilde{C}_A C_{A,0})}{\partial (\tilde{x} L)} - k (\tilde{C}_A C_{A,0}) = 0 \\ u C_{A,0} = \left[u (\tilde{C}_A C_{A,0}) - D \frac{d(\tilde{C}_A C_{A,0})}{d(\tilde{x} L)} \right]_{\tilde{x}=0} \\ \left[\frac{d(\tilde{C}_A C_{A,0})}{d(\tilde{x} L)} \right]_{x=L} = 0 \end{cases}$$
$$\Rightarrow \begin{cases} \alpha \frac{\partial^2 \tilde{C}_A}{\partial \tilde{x}^2} - \frac{\partial \tilde{C}_A}{\partial \tilde{x}} - \beta \tilde{C}_A = 0 & \alpha = \frac{D}{uL} \\ \left[\tilde{C}_A - \alpha \frac{d\tilde{C}_A}{d\tilde{x}} \right]_{\tilde{x}=0} = 1 & \beta = \frac{Lk}{u} \\ \left[\frac{d\tilde{C}_A}{d\tilde{x}} \right]_{x=L} = 0 \end{cases}$$

AD: Nondimensionalization

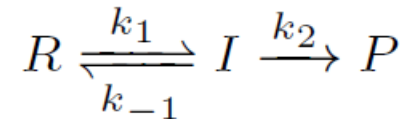
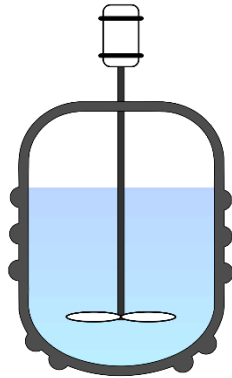
$$\alpha = \frac{D}{uL} = \frac{D (C_{A,0}/L) A}{uC_{A,0}A} \sim \frac{\text{axial dispersion rate for reactant A}}{\text{flow rate for reactant A}}$$

When AD is negligible, $\alpha \ll 1 \Rightarrow \begin{cases} -\tilde{C}'_A - \beta \tilde{C}_A = 0 \\ \tilde{C}_A(0) = 1 \end{cases} \quad \tilde{C}'_A(L) = 0$

When AD is not negligible (well mixing) $\alpha \gg 1 \Rightarrow \begin{cases} \tilde{C}''_A = 0 \\ -\alpha \tilde{C}'_A(0) = 1 \\ \Rightarrow \tilde{C}_A \text{ is constant} \end{cases} \quad \tilde{C}'_A(L) = 0$

Reaction in Batch Reactor

Main assumptions: (1) a well-mixed solution; (2) Constant T; (3) The fluid is incompressible



$$r_R = -k_1 C_R + k_{-1} C_I$$

$$r_I = k_1 C_R - k_{-1} C_I - k_2 C_I$$

$$r_P = k_2 C_I$$

$$\begin{cases} \frac{dC_R}{dt} = -k_1 C_R + k_{-1} C_I \\ \frac{dC_I}{dt} = k_1 C_R - k_{-1} C_I - k_2 C_I \\ \frac{dC_P}{dt} = k_2 C_I \end{cases} \text{ with initial conditions } \begin{cases} C_R(0) = C_{R,0} \\ C_I(0) = 0 \\ C_P(0) = 0 \end{cases}$$

Reaction in Batch Reactor: Nondimensionalization

$$\text{Let } \tilde{C}_* = \frac{C_*}{C_{R,0}} \quad \Rightarrow \quad \frac{d\tilde{C}_R}{dt} = -k_1\tilde{C}_R + k_{-1}\tilde{C}_I$$

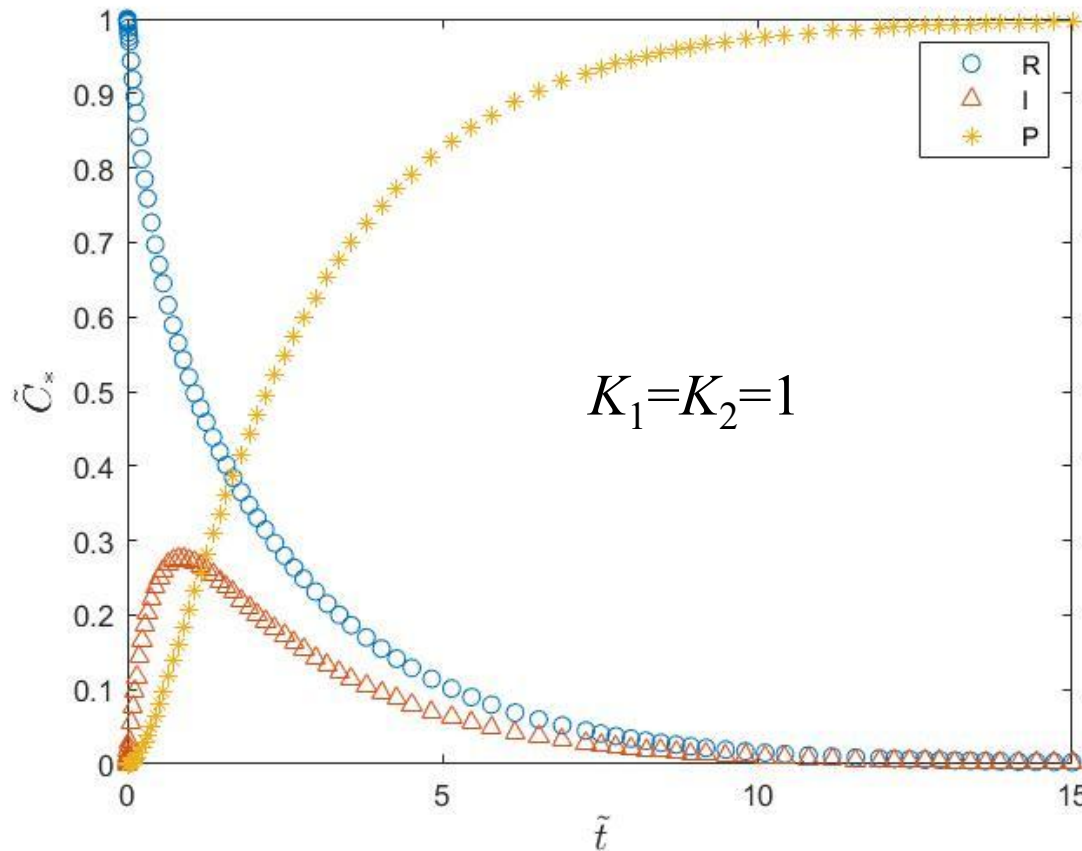
$$\Rightarrow \frac{d\tilde{C}_R}{k_1 dt} = -\tilde{C}_R + \frac{k_{-1}}{k_1}\tilde{C}_I$$

$$\Rightarrow \frac{d\tilde{C}_R}{d\tilde{t}} = -\tilde{C}_R + K_1\tilde{C}_I$$

$$\text{where } \tilde{t} = k_1 t \text{ and } K_1 = \frac{k_{-1}}{k_1}$$

$$\left\{ \begin{array}{l} \frac{dC_R}{dt} = -k_1 C_R + k_{-1} C_I \\ \frac{dC_I}{dt} = k_1 C_R - k_{-1} C_I - k_2 C_I \\ \frac{dC_P}{dt} = k_2 C_I \\ C_R(0) = C_{R,0} \\ C_I(0) = 0 \\ C_P(0) = 0 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \frac{d\tilde{C}_R}{d\tilde{t}} = -\tilde{C}_R + K_1\tilde{C}_I \\ \frac{d\tilde{C}_I}{d\tilde{t}} = \tilde{C}_R - K_1\tilde{C}_I - K_2\tilde{C}_I \\ \frac{d\tilde{C}_P}{d\tilde{t}} = K_2\tilde{C}_I \text{ for } K_2 = \frac{k_2}{k_1} \\ \tilde{C}_R(0) = 1 \\ \tilde{C}_I(0) = 0 \\ \tilde{C}_P(0) = 0 \end{array} \right.$$

Reaction in Batch Reactor: Plots



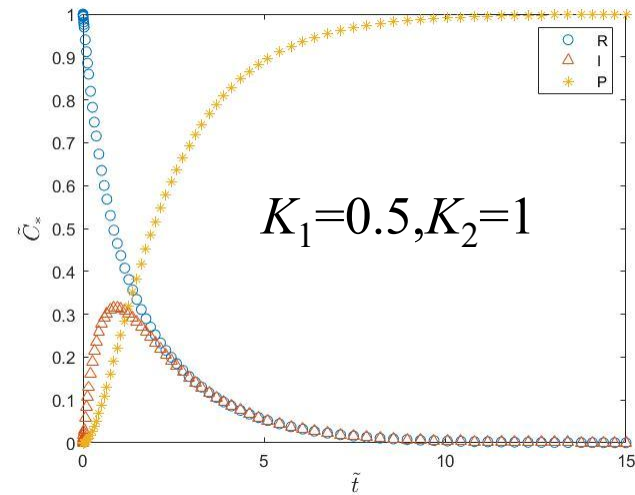
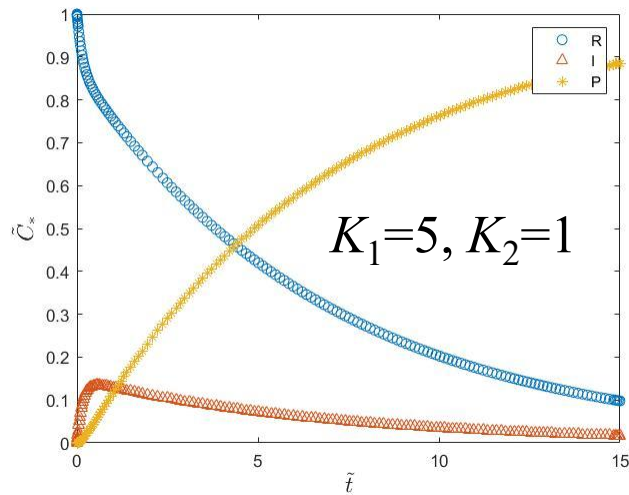
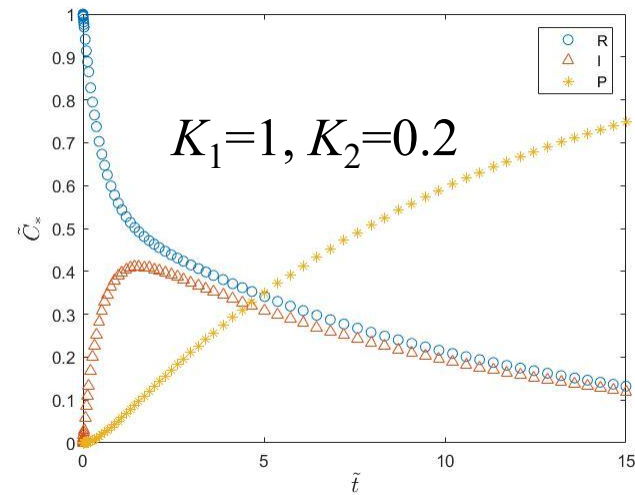
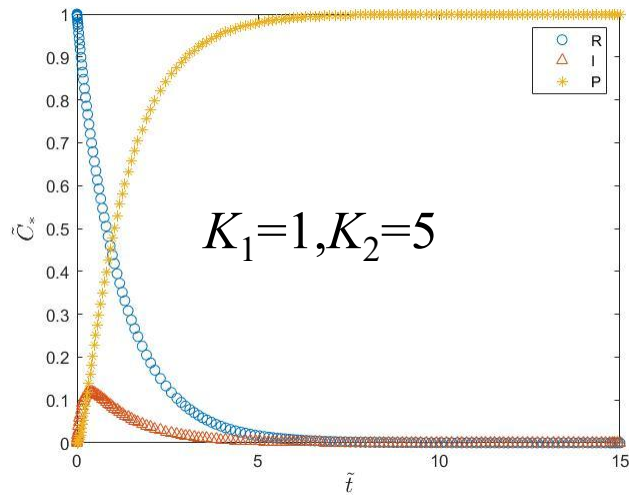
$$K_1 = \frac{k_{-1}}{k_1}$$

$$K_2 = \frac{k_2}{k_1}$$

What happens if K_1 and K_2 vary?

What insights can we derive from the graph?

Reaction in Batch Reactor: Plots

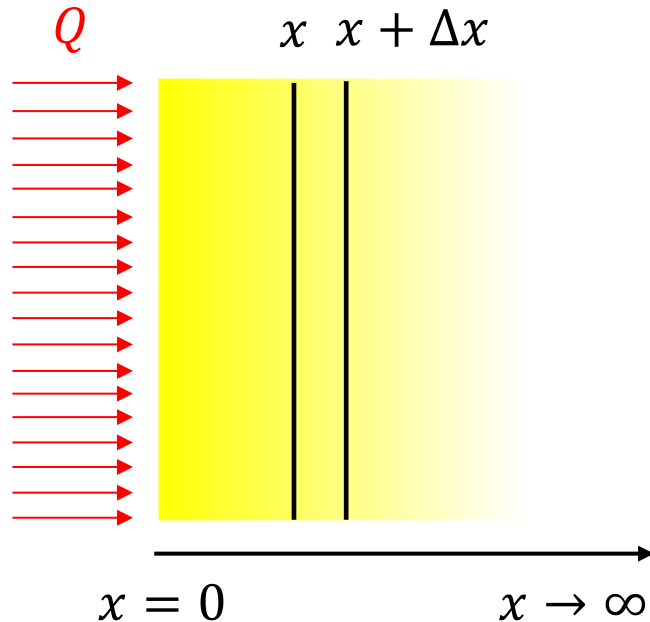


$$K_1 = \frac{k_{-1}}{k_1}$$

$$K_2 = \frac{k_2}{k_1}$$

Non-Steady-State Heat Transfer

Semi-infinite block



Fourier's Law: $q = -k \frac{dT}{dx}$

q : local heat flux density $\frac{\text{W}}{\text{m}^2}$

k : material conductivity $\frac{\text{W}}{\text{mK}}$

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial x^2} \right)$$

κ : thermal diffusivity

T : temperature

1. Derive the governing equation for heat transfer (heat equation)
2. Find appropriate BCs and ICs
3. Nondimensionalization

Derivation: Let's work together!

Nondimensionalization: Let's work together!

Homework Announcement (Week 2)

- **Release Date:** After the lecture on the same day
- **Submission:** Hard copy to be submitted upon arrival to class
- **Due Date:** Week 3 class – 2:00 PM, 11 Feb 2026
- **Grouping:** Clearly indicate all group members' names on the submission.
- **Start Early!** Don't wait until **10 Feb** — trust me, you'll regret it!